

DO INTERMEDIARIES MATTER FOR AGGREGATE ASSET PRICES?

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September 2018

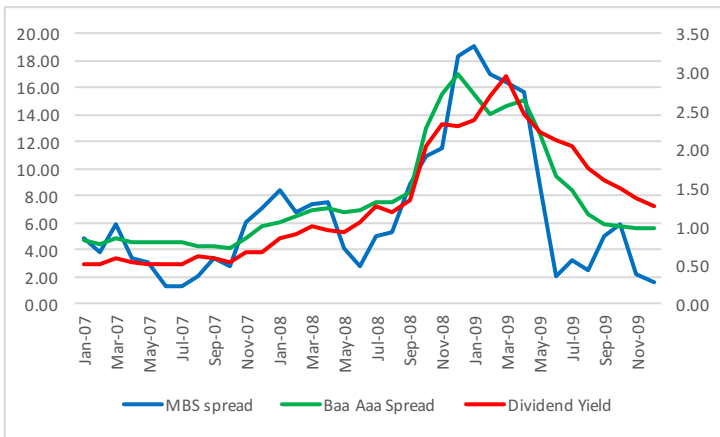
RESEARCH QUESTION

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EXAMPLE: 2008-09 FINANCIAL CRISIS



- Intermediary risk-bearing capacity was impaired
- But aggregate risk appetite also likely moved
 - habits, sentiment, etc

WHAT WE DO

Intermediary risk appetite matters more for assets that are difficult to directly invest in, household risk appetite matters less

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- 1 Overcomes identification issue of positive correlation of intermediary and household risk appetite
- 2 Theoretically justified
 - ▶ A model that nests the simple version of two main views
 - ▶ Existing “intermediary tests” do not get at the question
- 3 Across asset classes, we find:
 - ▶ Measures of financial sector health predict returns more strongly in asset classes that are difficult to invest in
 - ▶ Household measures have opposite pattern
 - ▶ Unrelated to *observable* variation in risk (vol, skewness, or beta)

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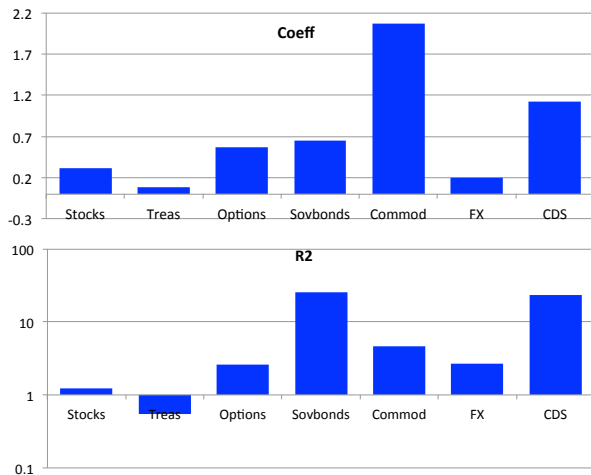
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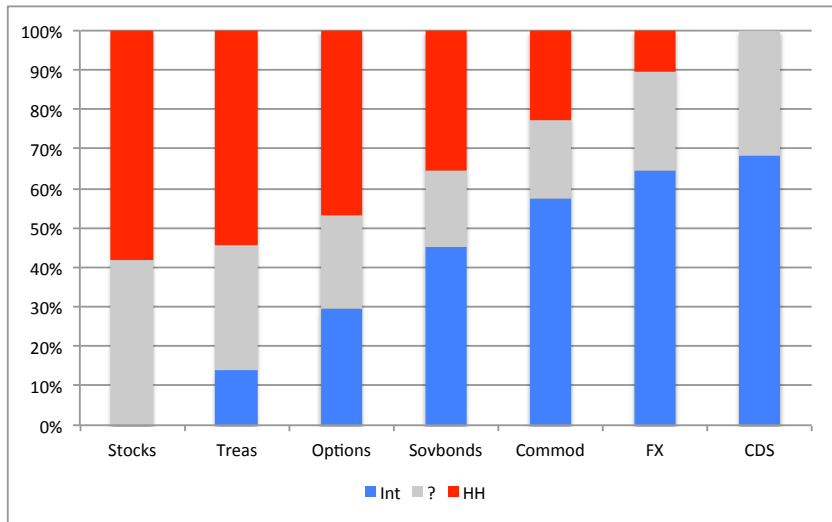
→ *Intermediaries and households have distinct, sizable effect on risk premia*

RESULTS

$$\tilde{r}_{i,t+1} = a_i + b_i \hat{\gamma}_{I,t} + \varepsilon_{i,t+1}$$



BOUNDS OF RISK PREMIA VOLATILITY



Use panel to provide lower bound of volatility due to each

LITERATURE

- **Aggregate** prices **consistent** with role of intermediaries: optimal decisions
 - ▶ Exposure to intermediary factor explains the cross-section of returns, e.g. Adrian Etula Muir (2014), He Kelly Manela (2017), intermediary balance sheet predicts returns, e.g. Haddad Sraer (2016)
- **Local** evidence that intermediaries **cause** changes in prices
 - ▶ Arbitrage opportunities directly related to intermediation regulatory constraints, e.g. Du Tepper Verdelhan (2017), Lewis, Longstaff, Petrasek (2017).
 - ▶ Price dislocations tied to risk bearing capacity (e.g., Duffie 2010, Siriwardane 2017)
 - ▶ Implications for aggregate prices less clear

BASIC IDEAS

- Time-varying risk appetite of intermediaries $\gamma_{I,t}$ and of household risk aversion $\gamma_{H,t}$

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- Question: does $\gamma_{I,t}$ play a role for risk premium?

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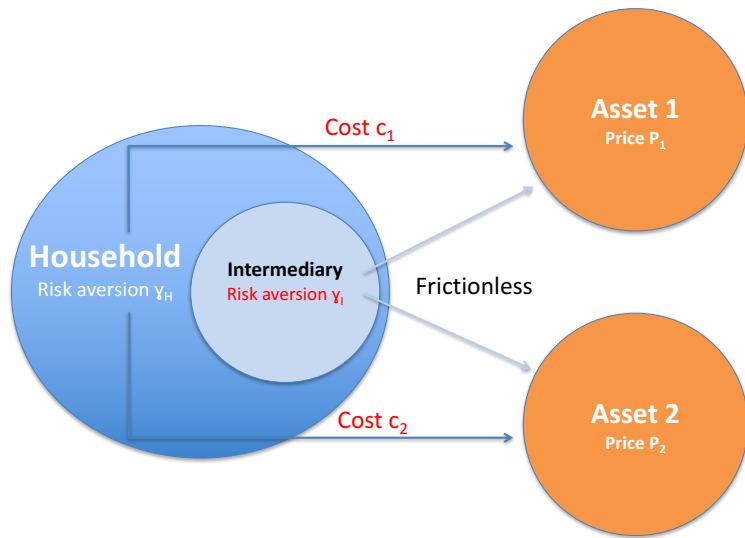
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 - ▶ Positively correlated with $\gamma_{I,t}$
 - ▶ Possibly positively correlated with $\gamma_{H,t}$
- Solution: Households face different frictions to invest in each asset class $\rightarrow \beta_{i,H}$ is (weakly) lower for more intermediated asset class

MODEL



- Two periods, N assets with payoffs $\mathcal{N}(\mu, \Sigma)$

SETUP

- **Household:** invest directly or through intermediary
 - ▶ CARA, risk aversion γ_H
 - ▶ Takes intermediary decisions as given
 - ▶ *Friction 1: Assets differ in their ease of access for direct investment (C)*

$$\begin{aligned} \max_{D_H} & (D_H + D_I)' (\mu - p) - \frac{\gamma_H}{2} (D_H + D_I)' \Sigma (D_H + D_I) \\ & - \frac{\gamma_H}{2} D_H' C D_H. \end{aligned}$$

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■ **Intermediary**

- ▶ *Friction 2: Intermediaries invest on behalf of household, but with different investment policies*
- ▶ CARA, risk aversion γ_I

$$\max_{D_I} D_I' (\mu - p) - \frac{\gamma_I}{2} D_I' \Sigma D_I.$$

INTERMEDIARY AND ASSET PRICES

$$\mu - p = \gamma_H \Sigma \left(\Sigma + \frac{1}{\gamma_I} C \right)^{-1} \left(\Sigma + \frac{1}{\gamma_H} C \right) S$$

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Proposition: The intermediary matters for asset prices, that is $\partial(\mu - p)/\partial\gamma_I \neq 0$, if and only if

$$\gamma_I \neq \gamma_H \quad \text{and} \quad C \neq 0$$

1. Imperfect substitution

- ▶ $C > 0 \Leftrightarrow \frac{\partial D_H}{\partial D_I} \neq -1$: Household doesn't undo intermediary decision.

2. Preference (mis)alignment

- ▶ $\gamma_I \neq \gamma_H$: Intermediary isn't a veil who acts perfectly on behalf of household

EXISTING TESTS

$$\mu - p = \gamma_H \Sigma \left(\Sigma + \frac{1}{\gamma_I} C \right)^{-1} \left(\Sigma + \frac{1}{\gamma_H} C \right) S$$

How can we test if intermediaries matter?

■ Existing test 1: Intermediary Euler equation

- ▶ Covariance with intermediary portfolio explains cross-section of expected returns
- ▶ Always holds regardless of C or γ_H : *optimality rather than equilibrium*

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■ Existing test 2: Time-series predictability

- ▶ High intermediary risk aversion γ_I predicts high returns
- ▶ **Identification challenge:** High household risk aversion γ_H also predicts high returns ... and γ_I may be correlated with unobservable γ_H

RELATIVE PREDICTABILITY

$$\beta_{i,I} = \frac{1}{\mu_i - p_i} \frac{\partial(\mu_i - p_i)}{\partial \log(\gamma_I)} = \frac{c_i}{\gamma_I \sigma_i^2 + c_i} \geq 0$$

$$\beta_{i,H} = \frac{1}{\mu_i - p_i} \frac{\partial(\mu_i - p_i)}{\partial \log(\gamma_H)} = \frac{\gamma_H \sigma_i^2}{\gamma_H \sigma_i^2 + c_i} > 0$$

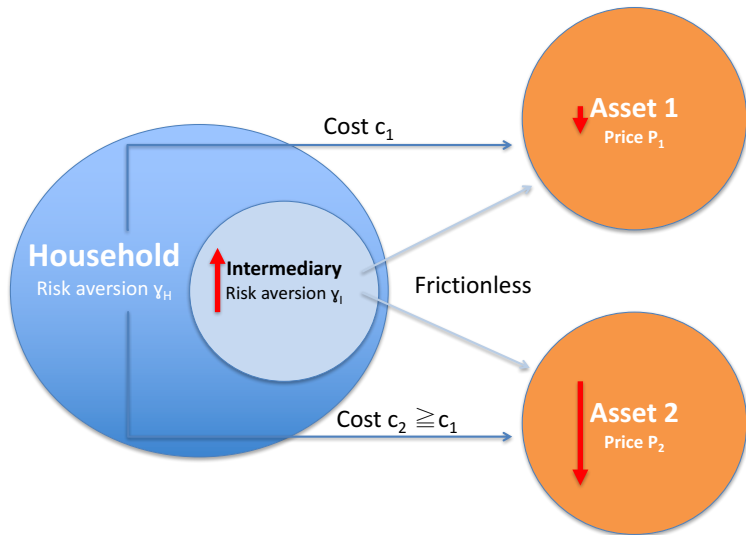
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- 1 Elasticity of risk premium to γ_I increasing in c_i
- 2 Elasticity to household risk aversion γ_H is decreasing in c_i

INTERMEDIARY RISK AVERSION



MORE GENERAL RESULT

- Specific to CARA normal 2 period model?

HH pricing kernel m^H . Frictionless factors driving risk premia:

$$E_t[r_{i,t+1}] = cov(-m_{t+1}^H, r_{i,t+1})$$

$$E_t[r_{i,t+1}] = \sigma_t(m_{t+1}^H)\sigma_t(r_{i,t+1})\rho_t(-m_{t+1}^H, r_{i,t+1})$$

Allow:

- Unobserved time-varying volatility of HH pricing kernel
- Control / check ex-post $\sigma_t(r_{i,t+1})$ doesn't drive result

Main assumption:

- $\rho_{i,t}$ doesn't change more for more intermediated assets when $\hat{\gamma}_{I,t}$ changes

RISK APPETITE PROXIES

- Measures of intermediary risk appetite $\hat{\gamma}_{I,t}$:
 - ▶ Adrian Etula Muir (2014) broker-dealer book leverage factor, He Kelly Manela (2016) factor market equity of primary-dealers
 - ★ Shown to proxy for health/distress of financial sector, linked to risk premiums
 - ▶ Take log annual change in variables as return predictors, standardize and average them together (show separately too)
- Measures of household risk aversion $\hat{\gamma}_{H,t}$
 - ▶ Habit: surplus consumption ratio from Cochrane (2017)
 - ▶ *cay* from Lettau Ludvigson (2001)
 - ▶ Consumer sentiment from Michigan Survey

RETURNS

Returns $r_{i,t+1}$:

- Stocks, Treasury bonds, Sovereign bonds, Options on stocks (straddle), Commodities, FX (carry trade), CDS
- **New:** Credit (high minus low yield corp return), MBS (Barclays hedged MBS index)
- Also look at returns to convertible bond arb, fixed income arb, other hedge fund strategies

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Normalization: different assets have different level of risk

- $r_{i,t+1}/E[r_i]$
- $r_{i,t+1}/\sigma[r_i]$

COST RANKINGS

- Create a ranking of direct investment costs c_i (low to high):

Stocks	Bonds	Options	Sov.	Com.	FX	CDS
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	Stocks	Bonds	Options	Sov.	Com.	FX	CDS
<i>FoF</i>	Stocks	Bonds		Sov Bonds			
<i>VaR</i>	Stocks	Bonds			Comm	FX	
<i>BIS</i>		Bonds	Options		Comm	FX	CDS

- Multiple sources

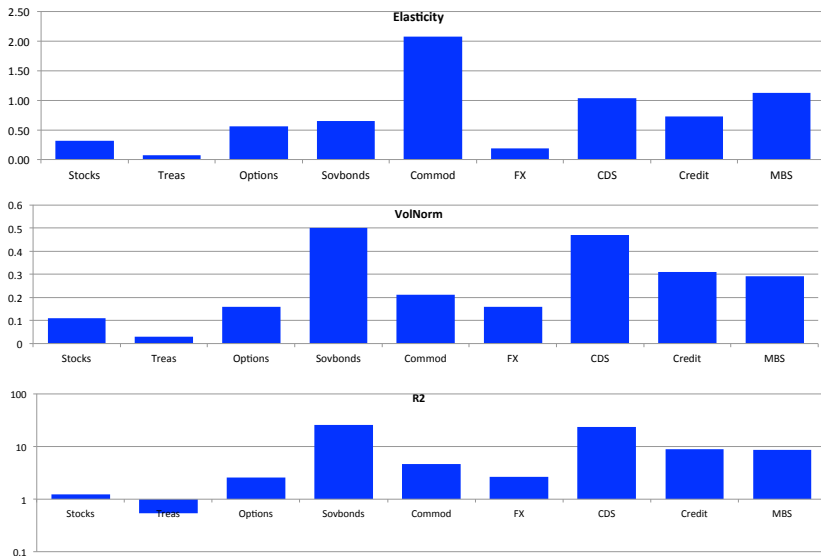
- ▶ *Flow of funds*: HH holdings / Total assets compared to broker dealers and other fin institutions
- ▶ *Value-at-Risk*: Take VaR for primary dealers (10K), normalize by asset class std dev, compare to size of each market
- ▶ *BIS* data on derivatives: Gross value, totals as well as accounted by fin institutions

INTERMEDIARIES AND RISK PREMIUM

$$r_{i,t+1}/E[r_i] = a_i + \textcolor{red}{b}_i \hat{\gamma}_{I,t} + \varepsilon_{i,t+1}$$

	(1) Stock	(2) Treas	(3) Opt	(4) Sov	(5) Comm	(6) FX	(7) CDS	(8) Cred	(9) MBS
γ_I	0.31 (0.26)	0.07 (0.15)	0.57* (0.10)	0.65*** (0.33)	2.07*** (0.79)	0.19* (0.11)	1.12** (0.45)	0.73** (0.33)	1.03** (0.46)
N	164	154	100	62	102	113	44	145	94
R^2	0.012	0.001	0.026	0.257	0.047	0.027	0.233	0.088	0.086

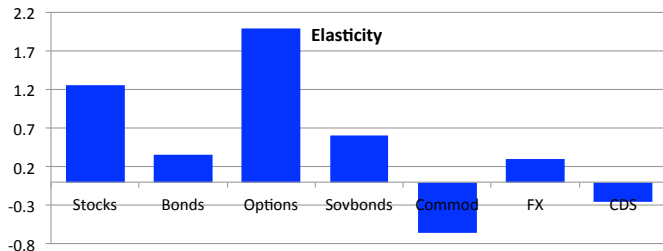
ALTERNATIVE SCALINGS



Vol norm: $r_{i,t+1}/\hat{\sigma}(r_{i,t+1}) = a_i + b_i\gamma_{I,t} + \varepsilon_{i,t+1}$

MECHANICAL?

Replace predictor with *cay* (HH risk aversion, similar for habit)



HH MEASURE

	(1) Stock	(2) Treas	(3) Opt	(4) Sov	(5) Comm	(6) FX	(7) CDS	(8) Cred	(9) MBS
γ_I	0.90 (0.50)	-0.02 (0.31)	0.58*** (0.16)	1.00** (0.45)	3.50* (1.92)	0.43** (0.20)	2.74*** (0.95)	1.38*** (0.48)	2.39*** (0.72)
γ_H	1.25*** (0.41)	0.26 (0.34)	0.25 (0.22)	0.61 (0.40)	-0.65 (2.04)	0.30 (0.26)	-0.26 (1.22)	-0.27 (0.31)	1.56* (0.85)
N	167	160	103	65	105	116	47	148	97
R^2	0.056	0.003	0.093	0.171	0.042	0.039	0.332	0.091	0.111

PANEL REGRESSIONS

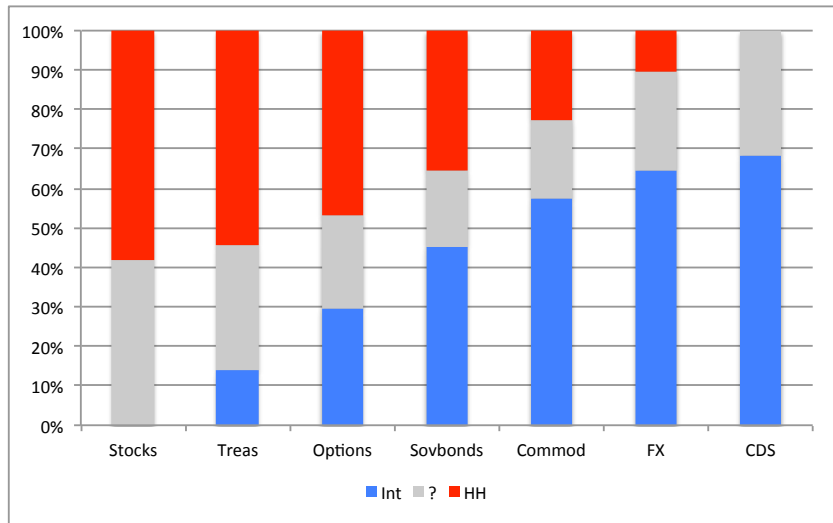
$$r_{i,t+1}/s_i = a_i + b_1^I \gamma_{I,t} + 1_{INT} b_2^I \gamma_{I,t} + b_1^H \gamma_{H,t} + 1_{INT} b_2^H \gamma_{H,t} + \varepsilon_{i,t}$$

- Panel regression with INT dummies for more intermediated assets (test if coeffs different)
- Add Campbell Cochrane habit (similar for *cay*)

TWO RISK PREMIUM CYCLES

$INT =$	$1_{\neq Stock/Bond}$	$1_{\neq Stock/Bond/Opt}$	Rank $\in [0,1]$			
γ_I	0.33* (0.18)	0.20 (0.18)	0.39** (0.20)	0.31 (0.19)	0.36* (0.21)	0.23 (0.21)
$INT \times \gamma_I$	0.56** (0.27)	0.76*** (0.28)	0.61** (0.30)	0.77** (0.34)	0.75** (0.37)	1.04** (0.41)
γ_H		0.41** (0.19)		0.29 (0.21)		0.40* (0.22)
$INT \times \gamma_H$		-0.61* (0.36)		-0.53 (0.37)		-0.85* (0.45)
N	730	730	730	730	730	730
R^2	0.0288	0.0335	0.0296	0.0330	0.0280	0.0320

BOUNDS OF RISK PREMIA VOLATILITY



Use panel to provide lower bound of volatility due to each

ROBUSTNESS

- Different samples: Table 8
 - ▶ Exclude crisis
 - ▶ More balanced panel: start post 1990
- Alternative measures of intermediary risk aversion: Tables 5-6
 - ▶ Use two measures separately
 - ▶ Use long-term changes in AEM/HKM or levels
 - ▶ Use Gilchrist Zachrajsek (2010) spread, health of intermediaries

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Next:

- Time-varying risk

TIME-VARYING RISK

- **Concern:** More intermediated assets become more risky exactly when intermediary health is poor... but nothing to do with intermediaries
- Measure and control for *observable* variation in risk

$$\ln(\sigma_{i,t+1}^2) = a_i + b_i \gamma_{I,t} + \varepsilon_{i,t+1}$$

	Mkt	Treas	Options	Sovereigns	Commodities	FX	CDS
γ_I	0.30*** (0.09)	-0.05 (0.10)	0.23*** (0.09)	0.20 (0.14)	0.35*** (0.10)	0.06 (0.11)	0.13 (0.21)
γ_H	0.12 (0.07)	0.28*** (0.10)	-0.02 (0.11)	0.27 (0.16)	0.20 (0.13)	-0.05 (0.07)	1.02*** (0.23)
N	164	154	100	62	102	113	44
R^2	0.139	0.055	0.0441	0.123	0.141	0.00818	0.431

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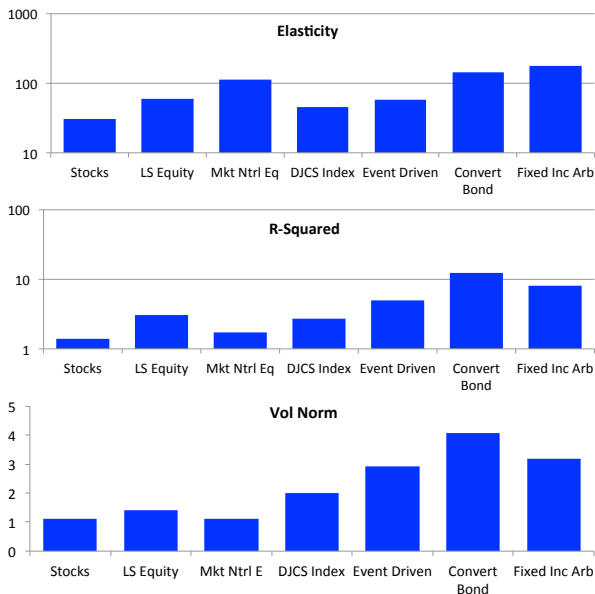
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- In addition: no differential effect for skewness, no difference when we control for time-varying betas, rolling vol, VIX, or other risk measures

ADDITIONAL EVIDENCE: HEDGE FUND RETURNS

- Convert bond arb and Merger arb (Mitchell and Pulvino (2001, 2012)): disruptions linked to capital scarcity, HF own 40+% of convertible bonds
- Fixed income arbitrage: Hu Pan Wang (2013)
- HF returns from DJCS: Equity LS, Mkt Neutral, Event Driven, Convert Bond, Fixed Income Arb, Total Index

HEDGE FUND RETURNS: INTERMEDIARY ($\hat{\gamma}_I$)



Note: pattern *not* there for γ_H

CONCLUSION

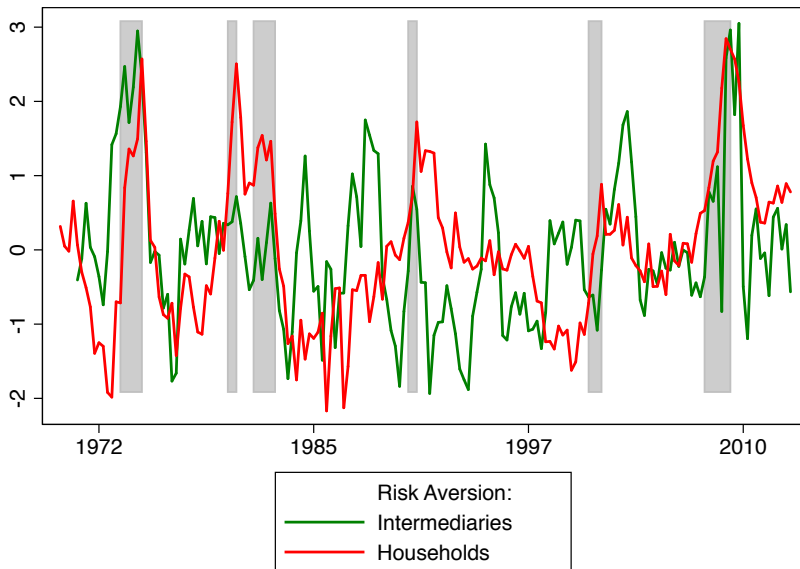
Do intermediaries matter for **aggregate asset prices?**

Yes, a lot. Households too.

- Intermediary risk appetite matters more for assets that are difficult to directly invest in
- Household appetite matters less
- Both results are specific signature of models with financial frictions.

APPENDIX

RISK APPETITE PROXIES



STATISTICAL PROPERTIES

Test if elasticity different?

$$r_{i,t+1}/\overline{(r_{i,t+1})} - r_{stock,t+1}/\overline{(r_{stock,t+1})} = a_i + b_i\gamma_{I,t} + \varepsilon_{i,t+1}$$

Elasticity Difference						
	Bonds	Options	Sovereign	Commodity	FX	CDS
γ_I	0.26 (0.20)	-0.06 (0.15)	-0.01 (0.36)	-1.40 (0.85)	0.09 (0.28)	-0.08 (0.36)
N	145	100	62	102	113	44
R^2	0.016	0.003	0.002	0.028	0.050	0.002

- Instability of estimate in smaller sample: $\overline{r_{i,t+1}}$ hard to estimate, blows up std errors
- Elasticity “ideal” from theory, difficult test with smaller samples

VARIANCE NORMALIZATION

Variance norm more stable (easier to estimate than $E[r]$ in small sample)

$$r_{i,t+1}/\hat{\sigma}^2(r_{i,t+1}) - r_{stock,t+1}/\hat{\sigma}^2(r_{stock,t+1}) = a_i + b_i\gamma_{I,t} + \varepsilon_{i,t+1}$$

	Variance Normalization Difference					
	Bonds	Options	Sovereign	Commodity	FX	CDS
γ_I	-2.22*	-0.14	-3.11***	-0.87	-1.79**	-14.88**
	(1.21)	(0.20)	(1.13)	(0.68)	(0.76)	(6.66)
N	145	100	62	102	113	44
R^2	0.013	0.004	0.191	0.011	0.139	0.238

- Variance normalization less pure from theory (e.g., need to assume diagonal Σ) but more stable empirically in subsamples

INTERMEDIARY RISK AVERSION: HKM AND AEM

Rather than combine HKM AEM measures, here split separately

	Stocks	Bonds	Options	Sovereign	Commod	FX	CDS
Annual Changes							
γ_I^{AEM}	-0.42 (0.26)	-0.22* (0.12)	-0.90*** (0.26)	-0.50*** (0.15)	-3.44*** (0.58)	-0.26*** (0.08)	-0.79** (0.38)
γ_I^{HKM}	-0.04 (0.27)	-0.27 (0.18)	0.25 (0.37)	-0.39** (0.16)	1.12 (0.93)	0.01 (0.10)	-0.71* (0.39)
N	164	145	100	62	102	113	44
R^2	0.020	0.029	0.094	0.262	0.201	0.056	0.234

INTERMEDIARY RISK AVERSION: HKM AND AEM

Rather than combine HKM AEM measures, here split separately

	Stocks	Bonds	Options	Sovereign	Commod	FX	CDS
	Levels						
γ_I^{AEM}	-0.01 (0.39)	0.31 (0.20)	-1.00** (0.49)	-0.75* (0.39)	-1.75 (1.49)	-0.22* (0.12)	-0.80 (0.76)
γ_I^{HKM}	-0.59 (0.37)	-0.32 (0.22)	-0.45 (0.54)	-0.63*** (0.20)	-0.23 (1.52)	0.42*** (0.16)	-0.78 (0.49)
N	168	145	100	62	102	113	44
R^2	0.041	0.020	0.117	0.214	0.035	0.095	0.137

INTERMEDIARY RISK AVERSION: LEVELS

- Replace changes in log AEM / HKM with levels to proxy for γ_I
 - ▶ Most theories: level matters, but there are large trends
 - ▶ Follow Adrian Moench Shin (2010), Schularick Taylor (2012), Baron Xiong (2016) using 1-3 year changes

	Stocks	Bonds	Options	Sovereign	Commodities	FX	CDS
γ_I	-0.53** (0.22)	-0.01 (0.18)	-1.29*** (0.34)	-1.16*** (0.28)	-1.72* (0.89)	0.18 (0.13)	-1.40** (0.58)
N	168	145	100	62	102	113	44
R^2	0.033	0.000	0.110	0.212	0.027	0.020	0.137

INTERMEDIARY RISK AVERSION: GZ SPREAD

- Replace AEM / HKM with Gilchrist Zakrajsek (2012) excess bond premium spread
- ▶ GZ argue this captures health of intermediaries

	Stocks	Bonds	Options	Sovereign	Commodities	FX	CDS
<i>GZ</i>	-0.01 (0.28)	-6.14*** (1.09)	0.86 (0.84)	-3.10*** (1.01)	0.83 (1.05)	-0.38 (0.98)	-12.35* (4.09)
N	156	145	100	62	102	113	44
<i>R</i> ²	0.000	0.129	0.024	0.204	0.016	0.002	0.253

SUBSAMPLE: EXCLUDE CRISIS

Dropping the crisis (Panel A), Post 1990 only (Panel B)

Dropping 2007-2009							
	Stocks	Bonds	Options	Sovereign	Commodities	FX	CDS
γ_I	-0.22 (0.30)	-0.26 (0.17)	-0.49* (0.27)	-0.73*** (0.18)	-2.74*** (0.75)	-0.25** (0.11)	-0.90*** (0.15)
N	141	126	81	46	79	90	21
R^2	0.007	0.010	0.037	0.354	0.170	0.057	0.628

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SUBSAMPLE: POST 1990

	Post 1990						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Stocks	Bonds	Options	Sovereign	Commodities	FX	CDS
γ_I	-0.42 (0.30)	-0.41*** (0.10)	-0.42 (0.50)	-0.64*** (0.17)	-4.25** (1.98)	-0.23** (0.11)	-1.07*** (0.38)
N	84	80	80	62	84	84	44
R^2	0.025	0.163	0.008	0.254	0.038	0.035	0.231

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HOUSEHOLD RISK AVERSION: CONSUMER SENTIMENT

Proxy for $\gamma_{H,t}$ using consumer sentiment from Michigan survey

	Stocks	Bonds	Options	Sovereign	Commodity	FX	CDS
γ_I	-0.65 (0.57)	-0.51* (0.29)	-1.32* (0.73)	-1.17** (0.51)	-3.86** (1.92)	-0.55** (0.22)	-3.04*** (0.98)
γ_H	0.16 (0.55)	-0.10 (0.41)	-0.06 (0.84)	-0.26 (0.35)	-1.39 (2.59)	-0.47 (0.29)	-0.89 (1.03)
N	167	148	103	65	105	116	47
R^2	0.015	0.015	0.036	0.147	0.047	0.060	0.355